

11. Swap Two 8-bit Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center is as follows:

```
1 LDA 8050H
2 MOV B,A
3
4 LDA 8051H
5 MOV C,A
6
7 MOV A,B
8 STA 8051H
9
10 MOV A,C
11 STA 8050H
12
13 HLT
```

The registers on the left show the following values:

Register	Value
A	03
BC	02 03
DE	00 00
HL	00 00
PSW	00 00
PC	42 11
SP	FF FF
Int-Reg	00

The flags on the right show the following values:

Flag	Value
S	0
Z	0
AC	0
P	0
C	0

The memory window on the right shows the following data:

Address (Hex)	Address	Data
8050	32848	3
8051	32849	2
8052	32850	0
8053	32851	0
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

The I/O Ports window shows the following values:

Port	Value
0	- + 00

The Memory window shows the following values:

Address	Value
0	- + 00

The assembler message window at the bottom right shows the message: "Program assembled successfully".

12. 1's Complement of an 8-bit Number

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center is as follows:

```
1 LDA 8050H
2 CMA
3 STA 8051H
4 HLT
```

The registers on the left show the following values:

Register	Value
A	AA
BC	00 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 08
SP	FF FF
Int-Reg	00

The flags on the right show the following values:

Flag	Value
S	0
Z	0
AC	0
P	0
C	0

The memory window on the right shows the following data:

Address (Hex)	Address	Data
8050	32848	85
8051	32849	170
8052	32850	0
8053	32851	0
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

The I/O Ports window shows the following values:

Port	Value
0	- + 00

The Memory window shows the following values:

Address	Value
0	- + 00

The assembler message window at the bottom right shows the message: "Program assembled successfully".

13. 2's Complement of an 8-bit Number

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly window contains the following code:

```
1 LDA 8050H
2 CMA
3 INR A
4 STA 8051H
5 HLT
```

The registers window shows the following values:

Register	Value
A	DB
BC	00 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 09
SP	FF FF
Int-Reg	00

The flag window shows the following values:

Flag	Value
S	1
Z	0
AC	0
P	1
C	0

The memory window shows the following values:

Address (Hex)	Address	Data
8050	32848	37
8051	32849	219
8052	32850	0
8053	32851	0
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

The assembler message window shows the following message:

```
0 Program assembled successfully
```

14. Check Whether an 8-bit Number is Odd or Even

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly window contains the following code:

```
1 LDA 8050H
2 ANI 01H
3 JZ EVEN
4 MVI A, 01H
5 STA 8051H
6 HLT
7 EVEN: MVI A, 00H
8 STA 8051H
9 HLT
```

The registers window shows the following values:

Register	Value
A	00
BC	00 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

The flag window shows the following values:

Flag	Value
S	0
Z	1
AC	1
P	1
C	0

The memory window shows the following values:

Address (Hex)	Address	Data
8050	32848	36
8051	32849	0
8052	32850	0
8053	32851	0
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

The assembler message window shows the following message:

```
0 Program assembled successfully
```

15. Check Whether an 8-bit Number is Positive or Negative

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers Flag

Register	Value	Flag	Value
A	00	S	0
BC	00 00	Z	1
DE	00 00	AC	1
HL	00 00	P	1
PSW	00 00	C	0
PC	42 14		
SP	FF FF		
Int-Reg	00		

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00 Update Port Value

Memory

0 - + 00 Update Memory

Load me at

```
1 LDA 8050H
2 ANI 80H
3 JZ POSITIVE
4
5 MVI A, 01H
6 STA 8051H
7 HLT
8
9 POSITIVE: MVI A, 00H
10 STA 8051H
11 HLT
```

Start 8050h OK

Address (Hex)	Address	Data
8050	32848	69
8051	32849	0
8052	32850	0
8053	32851	0
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle